

Confidence Transport for Adaptive Experimentation: Theory and a Controlled Audit

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Meta

Proposed revision for author review.

Abstract

Model updates change the sensitivities used for allocation, while reward confidence may rely on sensitivities fixed when observations were collected. We give a finite-action confidence-transfer result under explicit pre-reward metric and solver certificates, yielding reward optimism and a regret bound on their joint validity event. Under fixed-reference and approximation-envelope assumptions, a corrected center supplies confidence without nonlinear optimizer convergence. A controlled nearly linear, 29-parameter simulation records simultaneous float64 confidence and optimism diagnostics, yet the evaluated regret bound is vacuous. Within each primary-horizon condition, enabling the analytic transport multiplier increases sample mean pseudo-regret relative to omitting it in an otherwise matched policy.

1 Adaptive model updates and confidence transport

An allocation policy might retrain an outcome model in batches, then use predicted rewards and uncertainty to choose an ad or ranking action. This illustration differs from our synthetic study. Learned reward models also appear in neural contextual bandits [2, 3]. Recomputing past sensitivities changes the Gram matrix. Collection-time confidence alone gives no upper bound using the recomputed matrix: an inverse-metric comparison for the current query is missing.

At round $t \leq T$, observe x_t , choose a_t from a fixed finite nonempty set $\mathcal{A}$, then observe r_t . Our confidence target is the conditional mean reward $\mu_t^*(a)$ used for allocation, not a treatment-effect estimand. Let $\mathcal{H}_t^-$ precede fresh round- t randomization Ω_t and $\mathcal{H}_t = \mathcal{H}_t^- \vee \sigma(\Omega_t)$ contain all pre-reward information. Queries $q_t(a) \in \mathbb{R}^d$ use fixed coordinates and dimension. The reference and exact current metrics, $\beta_t \geq 0$, and θ_t when used are $\mathcal{H}_t^-$ -measurable (predictable). All remaining score inputs and the selected action are finite and $\mathcal{H}_t$ -measurable, actionwise; each selected query is fixed before its reward. All validity events are measurable, and any claimed certificate probability is a supplied simultaneous bound.

For $\lambda, \sigma > 0$, with I the d -dimensional identity, define reference geometry and widths by

$$\bar{V}_1 = \lambda I, \quad \bar{V}_{t+1} = \bar{V}_t + \sigma^{-2} q_t(a_t) q_t(a_t)^\top, \quad \bar{s}_t(a)^2 = q_t(a)^\top \bar{V}_t^{-1} q_t(a). \quad (1)$$

Define cumulative pseudo-regret and frozen information gain, with $\gamma_0 = 0$, as

$$R_T = \sum_{t=1}^T \left[\max_{a \in \mathcal{A}} \mu_t^*(a) - \mu_t^*(a_t) \right], \quad \gamma_T = \log \frac{\det \bar{V}_{T+1}}{\det \bar{V}_1}. \quad (2)$$

Let V_t be the exact current metric and C_t the algorithmic operator, both $d \times d$ symmetric positive-definite (SPD). Set $s_t(a)^2 = q_t(a)^\top V_t^{-1} q_t(a)$ and $s_{C,t}(a)^2 = q_t(a)^\top C_t^{-1} q_t(a)$.

Theorem 1 (Finite-action confidence transfer). Under this protocol, suppose $\|q_t(a)\|_2 \leq G < \infty$. Define E_{conf} as the event that, simultaneously for $t \leq T$ and $a \in \mathcal{A}$,

$$|\mu_t^*(a) - m_t(a)| \leq \beta_t \bar{s}_t(a) + b_t(a), \quad b_t(a) \geq 0. \quad (3)$$

Let $\bar{D}_t \geq 0$, $0 < \kappa_{-,t} \leq \kappa_{+,t} < \infty$, and $1 \leq \alpha_t < \infty$ be finite pre-reward factors. Define E_{cert} as simultaneous validity, for every $t \leq T$, of the Loewner comparisons

$$e^{-\bar{D}_t} \bar{V}_t \leq V_t \leq e^{\bar{D}_t} \bar{V}_t, \quad \kappa_{-,t} V_t \leq C_t \leq \kappa_{+,t} V_t, \quad (4)$$

and, for a nonnegative pre-reward width map,

$$s_{C,t}(a) \leq \widehat{s}_t(a) \quad (a \in \mathcal{A}), \quad \widehat{s}_t(a_t) \leq \alpha_t s_{C,t}(a_t). \quad (5)$$

Use this same realized width map in the score and selector:

$$U_t(a) = m_t(a) + \beta_t e^{\bar{D}_t/2} \sqrt{\kappa_{+,t}} \widehat{s}_t(a) + b_t(a), \quad U_t(a_t) \geq \max_{a \in \mathcal{A}} U_t(a) - \xi_t, \quad \xi_t \geq 0. \quad (6)$$

Changing a scored width after selection requires rechecking the selector inequality. On $E_{\text{conf}} \cap E_{\text{cert}}$, $U_t(a) \geq \mu_t^*(a)$ for every action and round, and pathwise

$$\begin{aligned} R_T \leq & \sqrt{\left(\sigma^2 + \frac{G^2}{\lambda}\right) \gamma_T \sum_{t=1}^T \beta_t^2 \left[1 + \alpha_t e^{\bar{D}_t} \sqrt{\frac{\kappa_{+,t}}{\kappa_{-,t}}}\right]^2} \\ & + 2 \sum_{t=1}^T b_t(a_t) + \sum_{t=1}^T \xi_t. \end{aligned} \quad (7)$$

If $\Pr(E_{\text{conf}}) \geq 1 - \delta$ and $\Pr(E_{\text{cert}}) \geq 1 - \delta_{\text{cert}}$, these conclusions hold with probability at least $1 - \delta - \delta_{\text{cert}}$; independence is unnecessary.

Theorem 1 states the certificate composition directly. Classical Thompson geometry supplies endpoint and path-length constructions of its first metric comparison [4]. The corrected-center construction below supplies reference confidence without nonlinear optimizer convergence.

2 A corrected center without optimizer convergence

Use a differentiable model mean $\mu_\theta(x, a)$ in fixed parameter coordinates $\theta \in \Theta \subseteq \mathbb{R}^d$, with Θ a convex smoothness region. For $z = (x, a)$, define $q_\theta(z) = \nabla_\theta \mu_\theta(z)$, $q_t(a) = q_{\theta_t}(x_t, a)$, $z_s = (x_s, a_s)$, and $q_s := q_{\theta_s}(x_s, a_s)$. The representation path $\theta_t \in \Theta$ is predictable. Fix before data collection $\theta^\circ \in \Theta$, $\|\theta^\circ\|_2 \leq S$, with $\mu_t^*(a) = \mu^*(x_t, a)$ and $\mu^*(z) = \mu_{\theta^\circ}(z) + m^\circ(z)$. Observed rewards obey $r_t = \mu_t^*(a_t) + \eta_t$ and $\mathbb{E}[e^{u\eta_t} \mid \mathcal{H}_t] \leq e^{u^2 \sigma^2/2}$ for every real u .

Supply nonnegative pre-reward bounds $\epsilon_{\text{lin}}(s)$ on the absolute Taylor remainder of $\mu_{\theta^\circ}(z_s)$ about θ_s , and $\epsilon_{\text{mis}}(s)$ for $|m^\circ(z_s)|$. Define the current actionwise envelopes $\epsilon_{\text{lin},t}(a)$, $\epsilon_{\text{mis},t}(a)$ analogously about θ_t ; the past sum below is known before selection. Containment alone does not make these errors small. With access to the exact auxiliary frozen ridge estimator, set

$$y_s = r_s - \mu_{\theta_s}(z_s) + q_s^\top \theta_s, \quad \widehat{\theta}_t^{\text{lin}} = \bar{V}_t^{-1} \sigma^{-2} \sum_{s < t} q_s y_s, \quad (8)$$

$$m_t^{\text{corr}}(a) = \mu_{\theta_t}(x_t, a) + q_t(a)^\top (\widehat{\theta}_t^{\text{lin}} - \theta_t). \quad (9)$$

For $\delta \in (0, 1)$, self-normalization [1] and the Taylor envelopes give (3) with probability at least $1 - \delta$, simultaneously over rounds and actions, using $m_t = m_t^{\text{corr}}$,

$$\beta_t^{\text{corr}} = \sqrt{\gamma_{t-1} + 2 \log(1/\delta)} + \sqrt{\lambda} S + \frac{1}{\sigma} \sqrt{\sum_{s < t} [\epsilon_{\text{lin}}(s) + \epsilon_{\text{mis}}(s)]^2}, \quad b_t(a) = \epsilon_{\text{lin},t}(a) + \epsilon_{\text{mis},t}(a). \quad (10)$$

No nonlinear optimizer-convergence assumption is used. The exact current squared-loss generalized Gauss-Newton (GGN) metric is $V_t = \lambda I + \sigma^{-2} \sum_{s < t} q_{\theta_t}(z_s) q_{\theta_t}(z_s)^\top$.

3 Controlled audit

We ran four policies in a nearly linear scaled-tanh bandit with five actions and 29 parameters. For normalized features $\|\phi(z)\|_2 \leq 1$, the mean is $\mu_{\theta,W}(z) = \sqrt{W} \tanh(\phi(z)^\top \theta / \sqrt{W})$. Exact realizability holds with $\|\theta^\circ\|_2 \leq R = 1$; parameters

are projected to radius R , $\lambda = 1$, and Gaussian noise has $\sigma = 0.25$. For each horizon $T \in \{250, 500, 1000\}$ and target $D_{\text{target}} \in \{0.25, 0.5, 1, 2\}$, the predeclared rule sets

$$W = \left[\frac{4c_h R \sqrt{T-1}}{\sigma \sqrt{\lambda} D_{\text{target}}} \right]^2, \quad c_h = \frac{4}{3\sqrt{3}}, \quad L_g = \frac{c_h}{\sqrt{W}}, \quad Q_t = \sum_{s < t} \|\theta_t - \theta_s\|_2^2, \quad D_{Q,t} = \frac{2L_g \sqrt{Q_t}}{\sigma \sqrt{\lambda}}. \quad (11)$$

At $T = 1000$, the chosen scales make the model uniformly close to its linear counterpart on the stated domain. For $u = \phi(z)^\top \theta$, $|u| \leq 1$, and $f_W(u) = \sqrt{W} \tanh(u/\sqrt{W})$,

$$0 \leq 1 - \frac{f_W(u)}{u} \leq \frac{u^2}{3W} \leq \frac{1}{3W},$$

with $f_W(u)/u = 1$ by continuity at zero. The four conditionwise uniform upper bounds range from 1.375×10^{-7} to 8.798×10^{-6} . This quantifies the study’s near-linearity; it is not floating-point exactness or a restriction on what other transport certificates can achieve. Here L_g bounds the operator norm of the mean Hessian along replay segments. The target is a worst-case trust-region label, not realized path length or parameter displacement; W is a smoothness scale. Separate runs at each horizon and 50 evaluation seeds give 2,400 trajectories. Ten disjoint tuning seeds selected shared optimizer settings by mean squared prediction error. Policies share context and potential-noise streams but have their own histories, parameters, centers, radii, and metrics. All use dense float64 Cholesky solves and finite score maximization, with failure level $\delta = 0.05$. The exact-arithmetic primary instantiation has $C_t = V_t$, $\kappa_{-,t} = \kappa_{+,t} = \alpha_t = 1$, and $\xi_t = 0$.

The transport-endpoint policy is a dense endpoint-certificate comparator. It substitutes $d_{\text{Th},t} = \max_i |\log \lambda_i(V_t, \bar{V}_t)|$, where $V_t v_i = \lambda_i \bar{V}_t v_i$. This is the smallest nonnegative common exponent for the symmetric all-direction sandwich in (4) [4], not an optimum over action-specific or statistical certificates. Both matrices are learner-known before reward observation. The certificate is valid in exact arithmetic; its float64 evaluation is not a verified enclosure. Every method’s runner computes these eigenvalues, so reported runtime does not isolate the endpoint policy’s incremental cost. Each policy scores $m_t^{\text{corr}}(a) + \beta_t^{\text{corr}} w_t(a) + b_t(a)$ on its own history:

$$\begin{array}{ll} \text{transport Hessian:} & w_t = e^{D_{Q,t}/2} s_t, \\ \text{frozen reference:} & w_t = \bar{s}_t, \end{array} \quad \begin{array}{ll} \text{transport endpoint:} & w_t = e^{d_{\text{Th},t}/2} s_t, \\ \text{naive current:} & w_t = s_t. \end{array}$$

The primary policy uses $\bar{D}_t = D_{Q,t}$; “Hessian” refers to the envelope argument, not its GGN width. Frozen reference applies the current query to the collection-time Gram; naive current omits transport. Let $\mathcal{D}_{\text{selected},t}$ be the exact Thompson–Finsler length of the SPD path obtained by interpolating each replay parameter from θ_s to θ_t . Analytically, $d_{\text{Th},t} \leq \mathcal{D}_{\text{selected},t} \leq D_{Q,t}$. $\mathcal{D}_{\text{quad},t}$ estimates this integral using 32-point Gauss–Legendre quadrature at the declared checkpoints; unenclosed quadrature cannot replace the exact integral in that inequality.

D_{target}	0.25	0.5	1	2
Transport Hessian mean R_{1000}	81.05	82.44	87.04	93.99
Transport endpoint mean R_{1000}	79.05	79.49	80.06	81.89
Frozen reference mean R_{1000}	79.09	80.13	80.25	81.67
Naive current mean R_{1000}	79.00	79.49	80.10	81.88
Median $D_Q/\mathcal{D}_{\text{quad}}$	1.88×10^6	9.77×10^5	4.82×10^5	2.51×10^5
Median trajectory-maximum d_{Th}	1.31×10^{-7}	5.23×10^{-7}	2.10×10^{-6}	8.74×10^{-6}
Median $e^{D_Q/2}$	1.02	1.04	1.09	1.18

Table 1: Primary horizon $T = 1000$. Regret means average 50 trajectories. The last three rows concern the primary policy: path ratios pool eligible seed-checkpoint pairs with $\mathcal{D}_{\text{quad}} > 10^{-12}$; endpoint distances are maximized within each trajectory before taking the median; inflation medians pool seeds and all rounds, not terminal rounds or run maxima.

Within each fixed $T = 1000$ condition, enabling the analytic transport multiplier in the otherwise matched current-width policy increased sample mean cumulative pseudo-regret relative to omitting it (Table 1). The policies use the same state-to-score and update rules apart from this multiplier, with common context and potential-noise streams; their subsequent histories may diverge. This descriptive total-policy contrast does not identify a fixed-history mechanism, per-seed dominance, or a cross-condition dose response, since D_{target} also changes W . In each of the 12 horizon-target cells, 50/50 primary trajectories passed both all-action, all-round reference-confidence and optimism diagnostics, with cellwise exact 95% Clopper–Pearson interval [92.9%, 100%]. Base seeds are reused across cells; no deterministic audit

failed. These tolerance-based float64 checks and quadrature values are diagnostics, not verified enclosures. Across the 12 primary cells, medians of per-trajectory ratios of the numerical right-hand side (RHS) of (7) to pseudo-regret range from 126.7 to 155.7. Every primary trajectory’s terminal RHS exceeds $2T$, the pseudo-regret cap from $|\mu^*(z)| \leq 1$, despite unbounded Gaussian observations.

4 Implications for adaptive experimentation

The theorem supplies reward optimism and a regret bound on the joint confidence and certificate event; certificate validity alone does not ensure useful allocation. In this nearly linear simulation, the analytic multiplier increased sample mean pseudo-regret relative to its omission within each fixed condition. This does not establish a uniform policy ordering, a cross-condition dose response, or treatment-effect inference after adaptive assignment.

References

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